

A Literature Review on Electrodynamic Microphones

Bharadwaj Koorapati¹, Nitin Mediseti², Krishna Gaadha³, Romikumar Rameshbhai Kukadiya⁴, Akhil Kumar Dodla⁵, and Ashish Uriviti⁶

^{1,2,3,4,5,6}**Student, M.Eng. Mechatronic and Cyberphysical Systems, Technische Hochschule Deggendorf, Germany**

Advisor: Prof. Dr. Dmitry Rychkov

Department of Mechanical Engineering and Mechatronics, Technische Hochschule Deggendorf, Germany

Abstract

Electrodynamic microphones are known for their durability and efficiency in high pressure sound situations and continue to serve an important role even though they might not be as sensitive as condenser microphones. Electrodynamic microphones differ from condenser and ribbon microphones in their construction and application. While dynamic microphones operate on the principle of electromagnetic induction, where sound waves move a diaphragm within a magnetic field to induce an electrical current, condenser microphones use a capacitor-like diaphragm and require external power for higher sensitivity and a broader frequency response. Ribbon microphones, on the other hand, capture sound through a thin ribbon placed in a magnetic field, offering a smooth, natural sound but are more fragile. Ultimately, they may be more vulnerable to handling noise, heavier, more expensive, and occasionally require additional power. These devices continue to be important as microphone technology advances, offering audio enthusiasts and professionals an efficient and helpful way to record loud and clear sounds for a wide range of applications. Recent advancements in microphone technology, such as MEMS electrodynamic microphones, have addressed miniaturization challenges while improving performance and efficiency. Additionally, research into nonlinearities in electrodynamic microphones highlights areas for improvement in design and performance. This paper explores the working principle, advantages/disadvantages and applications of electrodynamic microphones.

Key words: Electrodynamic, Faraday's Law, Diaphragm, Electromagnetic Induction, Acoustic signals, Transducer, Lorentz Force

1 Introduction

Electrodynamic microphones have played a substantial role in sound recording technology over the years. They bring a good balance between sensitivity and frequency response, coupled with durability. Even though digital audio technology has grown over the years, electrodynamic microphones are still much in use in professional and consumer areas, like music production and podcasting. The enduring appeal of electrodynamic microphones stems from their ability to capture the nuances and subtleties of sound with remarkable accuracy. Their robust construction and relatively simple design make them reliable and long-lasting. In the current era of digital audio, where software and digital signal processing play a significant role in shaping sound, the importance of high-quality analog input devices like electrodynamic microphones remains undiminished. These microphones provide a crucial first step in the audio recording chain, capturing the original sound signal with minimal coloration or distortion. This paper delves into the fundamental principles of electrodynamic microphone operation and the characteristic differences between different types of microphones. By analysing the design choices and technological innovations in modern electrodynamic microphones, and aim to shed light on the reasons for their continued popularity and explore potential avenues for future development.

2 Working Principle

An electrodynamic microphone, also known as dynamic microphone, operates based on Faraday's Law of Electromagnetic Induction. This principle states that when a conductor moves within a magnetic field, an electric current is induced in the conductor. This principle enables the conversion of sound waves into an electrical audio signal.

2.1 Key components

An electrodynamic microphone typically consists of three main components: **Figure 2**

1. Diaphragm:

A thin, flexible membrane that vibrates in response to sound waves. It is typically made of materials such as Mylar or other polymers.

2. Coil (conductor):

A small lightweight coil of wire attached to the diaphragm, which moves within a magnetic field when the diaphragm vibrates.

3. Permanent Magnet:

A magnet that creates a stable magnetic field around the coil. Neodymium, alnico, or ferrite are common materials used as a permanent magnet.

4. Enclosure (Housing):

A casing that protects the internal components and provides structural support. It also often incorporates acoustic filters or baffles to control sound pickup.

5. Output Terminals:

These carry the induced electrical signal to an external device, such as an amplifier or recording device. [5]



Figure 1: Images of the Electrodynamic Microphones [1]

2.2 The Process of Sound-to-Electric Signal Conversion.

The functioning of an electrodynamic microphone can be broken down into the following steps: **Figure 2**

Step 1: Sound wave capturing

Sound waves in the environment, which are variations in air pressure, reach the diaphragm of the microphone. The diaphragm vibrates in response to these sound waves, moving back and forth in proportion to the intensity and frequency of the sound.

Step 2: Diaphragm and Coil Motion

The coil, which is attached to the diaphragm, moves within the magnetic field created by the permanent magnet. This motion of the diaphragm, and hence the coil attached to it, is proportional to the sound wave's frequency (pitch) and amplitude (loudness).

Step 3: Induction of Electrical Current:

The back-and-forth movement of the coil through magnetic field of the permanent magnet, induces an electric current in the coil according to Faraday's Law of Induction. The direction of the current changes as the coil moves in and out of the magnetic field, producing an alternating current (AC) signal. The magnitude of the induced current corresponds to amplitude of the sound wave, and the frequency of the current is same as that of the sound wave's frequency.

Step 4: Electrical Signal Output

The induced electrical signal is carried from the coil to the microphone's output terminals.

This signal is weak. It can then be sent to the preamplifier or any audio interface for further amplification and processing.

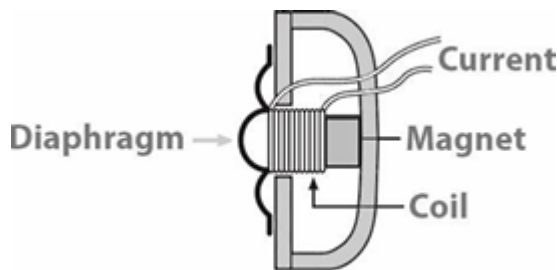


Figure 2: A schematic of an Electrodynamic Microphone [2]

future breakthroughs. In 1856, Italian inventor Antonio Meucci developed a dynamic microphone based on the generation of electric current by moving a coil of wire to various depths in a magnetic field. This method of modulation was also the most enduring method for the technology of the telephone as well. Speaking of his device, Meucci wrote in 1857, "It consists of a vibrating diaphragm and an electrified magnet with a spiral wire that wraps around it. The vibrating diaphragm alters the current of the magnet. These alterations of current, transmitted to the other end of the wire, create analogous vibrations of the receiving diaphragm and reproduce the word." [3] In 1861, Johann Philipp Reis (1834-1874), a German physicist, devised a "sound transmitter" that employed a metallic strip resting on a membrane to create intermittent contact in an electrical circuit as the membrane vibrated. It was Reis's theory that, as the membrane vibrated, the metal point bounced up and down 'producing intermittent contact and thus a varying current synchronous with the vibrations. However, this design was not refined enough for intelligible speech transmission. On March 10, 1876 Alexander Graham Bell employed a very similar transmitter design for the first transmission of intelligible speech over a rudimentary telephone system. Later on in 1878, David Edward Hughes paved a way forward in the communication industry with the carbon granule microphone, coining the term "microphone". Following his work, Thomas Alva Edison refined the carbon microphone which resulted in a carbon-button transmitter. The 1930s saw Harry

3 Literature Review

3.1 History of Microphones

The microphone has undergone a remarkable evolution since its inception. This section traces the chronological development of microphone technology, highlighting key innovations and advancements. The earliest attempts to capture and transmit sound electrically were rudimentary but laid the groundwork for

Olson’s development of the ribbon microphone, offering unparalleled natural sound quality and Eugen Beyer’s 1958 ”short diaphragm” ribbon microphone enhanced durability and reduced size. Sennheiser’s 1960s RF condenser microphones addressed humidity sensitivity, a major breakthrough for recording. In the modern scenario Soundfield microphones are widely used which enable ambisound surround sound recording which is essential for immersive audio. [4]

3.2 Recent Advancements in Electrodynamic Microphone Technology

Through the years, various advancements have shaped the way microphones fit into the current scenario. Most of the modern microphones use neodymium-iron-boron (NdFeB) magnets, which have excellent magnetic strength, allowing compact and lightweight microphones to be designed without sacrificing sensitivity. For the diaphragm, there are frequent uses of composite polymers like polypropylene for their durability and acoustic attributes. Recent research has brought graphene membranes into the picture of microphone design. With its ultra-lightweight and highly sensitive characteristics, Graphene allows for a broader frequency response and lower distortion, showing great promise as a material for next-generation microphones. Optical sensing materials and laser-sensitive polymers are also being integrated into high-performance microphones to achieve precise vibration detection. Researchers are also looking at bio-inspired materials, such as artificial spider silk, which have extraordinary sensitivity and flexibility. Those new concepts have expanded what electrodynamic microphones can achieve, enabling them to be used in studio recording, wearable devices, and modern audio technology.

3.3 A MEMS-Based Shifted Membrane Electrodynamic Microsensor for Microphone Applications

The advancement of Microelectromechanical Systems (MEMS) has revolutionized microphone technology, offering compact, energy-efficient, and high-

performance solutions for diverse applications. MEMS microphones operate on capacitive, piezoelectric, or piezoresistive principles to convert acoustic signals into electrical outputs. Capacitive MEMS microphones dominate the market due to their high sensitivity and broad frequency response. Innovations in membrane design and material optimization, such as using silicon, silicon nitride, and polymer composites, have enhanced their sensitivity and minimized noise. Similarly, piezoelectric MEMS microphones, which utilize materials like lead zirconate titanate (PZT), offer immunity to environmental noise, making them ideal for challenging environments. Recent research has explored the concept of shifted membrane designs to improve MEMS microphones. These designs enhance displacement sensitivity, damping, and linearity by altering the conventional planar membrane structure. In particular, electrodynamic MEMS microsensors, based on Lorentz force principles, demonstrate robust and linear acoustic responses. Studies have shown that integrating shifted membranes in electrodynamic designs significantly improves directional sensitivity and bandwidth, making them suitable for advanced applications like spatial audio capture and beamforming. MEMS-based shifted membrane electrodynamic microsensors face challenges, including fabrication complexity, material limitations, and environmental stability. Precise alignment of shifted membranes, development of optimal materials, and ensuring consistent performance across varying conditions remain active research areas. Addressing these challenges through advancements in nanofabrication and simulation techniques is critical for integrating these microsensors into next-generation acoustic systems. The continued evolution of MEMS technology promises to deliver efficient, compact, and high-performing solutions for the growing demands of modern audio applications. [7]

3.4 MEMS Electrodynamic microphone

The capacitive microphone is considered to be the most common type among silicon microphones, because of its high sensitivity (mV/Pa), large bandwidth and its low noise level. On the other hand,

piezoresistive microphones are robust, but suffer from low sensitivity and the piezoresistive material can suffer from thermal degradation due to Joule heating effects. The piezoelectric microphone is very common in aero acoustic applications, but also suffers from low sensitivity and low bandwidth. Lastly optical microphones drawbacks reside in the requirement of stable optical reference and encapsulation of all systems components such as light sources, optical sensor and photodetector, which should be accurately aligned and positioned. Electrodynamic microphone is very common in its traditional form of mm-size. They are characterized by low built-in noise and low internal impedance which makes them robust, relatively inexpensive and resistant to moisture. This, coupled with their potentially high gain (no preamplifier is required) makes them ideal for on-stage use. Nevertheless, some classic dynamic microphones still suffer from low sensitivity because of the slow vibration velocity of a heavy diaphragm upon which a spiral coil is attached, which makes the whole device quite bulky for integration. Some attempts to manufacture a miniaturized electrodynamic microphone were conducted. It combines a diaphragm with coils manufactured using MEMS technology (and hence, MEMS electrodynamic microphone), and a macro-magnet embedded in the external package. [6]

3.5 Advancing the Understanding of Nonlinearities in Electrodynamic Microphones

Electrodynamic microphones are widely used in audio recording and sound engineering due to their robust performance, durability, broad frequency response, sound reproduction quality and reliability. However, nonlinearities in their operation can introduce distortions that affect audio fidelity. Nonlinearities are typically caused by material limitations, mechanical constraints, and the inherent physics of sound wave conversion into electrical signals. Understanding and mitigating these nonlinear effects is crucial for high-precision applications, such as professional audio recording, scientific measurement, and telecommunication. Previous studies have primar-

ily addressed the linear characteristics of electrodynamic microphones, including sensitivity, impedance, and frequency response. Research into their nonlinear behaviour is less extensive but growing. This gap is critical because nonlinear distortions are more pronounced at higher sound pressure levels or extreme operational conditions, common in professional and industrial applications. Studies often employ controlled environments with calibrated sound sources and advanced measurement equipment to isolate and quantify these nonlinear effects. Experimental investigations into microphone nonlinearities typically involve subjecting the device to varying acoustic stimuli and measuring the output signal for distortion or other anomalies. These measurements help define the threshold beyond which the microphone's performance deviates from linearity. This gap underscores the need for comprehensive studies that bridge experimental data with theoretical modelling, thereby offering actionable insights for improving microphone design and functionality. This research adds to the field by providing experimental data on the nonlinear performance of electrodynamic microphones. By leveraging advanced experimental setups and precise measurement techniques, it provides detailed data on how these devices perform under varying acoustic loads. The findings contribute to a deeper understanding of the mechanisms driving nonlinearities and offer guidance for enhancing the design and application of electrodynamic microphones in high-demand scenarios. [8]

4 Comparative Analysis

Electrodynamic microphones are widely used for their durability, affordability, and good sound quality. **Table 1** is a comparative analysis of their key features against other types of microphones, such as condenser and ribbon microphones.

Electrodynamic	Condenser	Ribbon
Exceptionally robust; can withstand high SPL and extreme conditions.	Delicate due to sensitive components.	Can be damaged by air bursts or mishandling.
Most affordable due to simple construction.	Higher cost due to complexity.	More expensive with premium features.
Less sensitive; ideal for loud sound sources like drums and amplifiers.	Highly sensitive; captures subtle details but prone to ambient noise.	Moderate sensitivity with smooth, natural tones.
Limited high-frequency response; great for live performances and midrange instruments.	Broad frequency range; captures fine details in vocals and acoustic instruments.	Warm tone; excels in mid to low frequencies.
Ideal for live sound, podcasting, and high-SPL sources (e.g., snare drums, guitar cabinets).	Preferred in studios for vocals, strings, and detailed recordings.	Best for orchestras, horns, and voiceovers in studios.
Excellent isolation with directional polar patterns (e.g., cardioid).	More prone to picking up ambient noise.	Similar to condenser but with smoother response.
No external power needed.	Requires external power.	May need external power in modern designs.

Table 1: Comparison of Electrodynamic, Condenser, and Ribbon Microphones

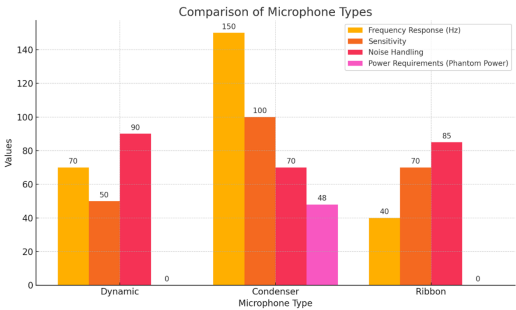


Figure 3: Comparison of Microphone types

4.1 Differences in Frequency Response Between Electrodynamic and Condenser Microphone

4.1.1 Dynamic Microphones

Dynamic Microphones generally exhibit a more colored frequency response due to their limitations of diaphragm and coil assembly. Their heavier diaphragms struggle to produce high frequencies, because, it is difficult for short-wavelength and high-frequency sounds to move the diaphragm. **Figure 4**

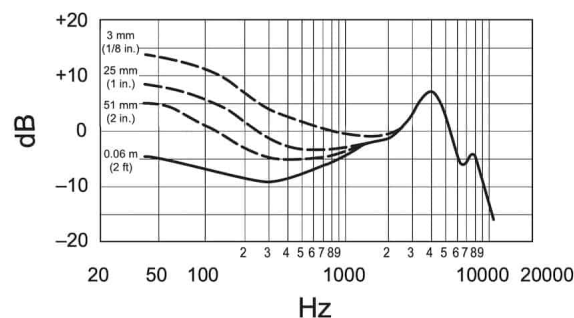


Figure 4: Frequency vs Response of a Dynamic Microphone [2]

4.1.2 Condenser Microphones

Condenser Microphones are known for their flatter and more extended frequency responses. These microphones are categorized into three types: Large-diaphragm condensers, which offer wide frequency coverage with a slight boost in the upper range **Figure 5**

4.1.3 Ribbon microphones

Ribbon Microphones on the other hand, offer a natural roll-off in high frequencies. With their loosely tensioned diaphragms, resonant frequencies fall below the audible range, and their high-frequency response, results in gradual and smooth roll-off.

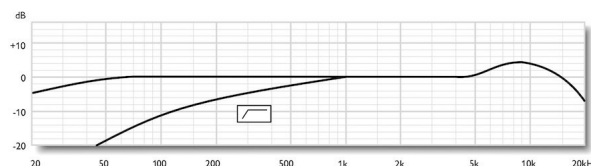


Figure 5: Frequency vs Response of a Condenser Microphone [2]

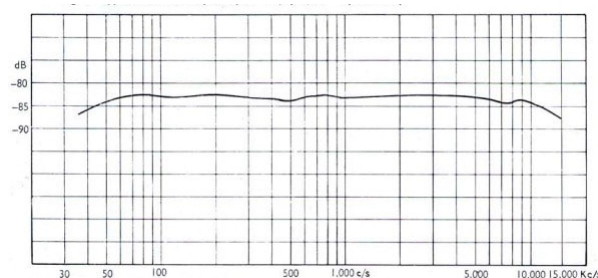


Figure 6: Frequency vs Response of a Ribbon Microphone [2]

5 Applications

1. **Live Sound Reinforcement:** Due to their ability to withstand high sound pressure levels without distortion, these systems are generally preferred for live performances, including concerts and events hence perfect for enhancing the sounds of different instruments and singers' voices.
2. **Recording Studios:** They are a mainstay in professional studio settings due to their outstanding sound capture capabilities and capacity to separate background noise. Because they guarantee excellent sound reproduction, they are particularly well-liked for recording singers and instruments like guitars and drums.
3. **Broadcasting:** These systems are widely utilized in television and radio studios because to their dependability and adaptability. They are

perfect for voiceovers, interviews, and presentations because of their versatility in a range of acoustic settings.

4. **Public Address Systems:** These systems are ideal for speeches, seminars, and public gatherings when excellent voice projection is essential because they are made to reduce handling noise.
5. **Telecommunications:** Although they are occasionally used in portable voice communication devices, they provide a good balance between sound quality and durability, which makes them appropriate for effective and useful communication.

6 Conclusion

Electrodynamic microphones remain essential in the audio industry due to their resilience, cost-effectiveness, and versatility. Their durability and capacity to handle high sound pressure make them invaluable in live performances, broadcasting, and various recording applications. Despite the rise of other microphone technologies such as condenser and ribbon microphones, electrodynamic microphones remain highly valued for their ability to capture high-pressure sound sources, making them ideal for live sound reinforcement, music production, broadcasting, and telecommunications. The principles of electromagnetic induction that drive their operation have stood the test of time, and their simplicity and reliability are key factors contributing to their enduring popularity. The simplicity of their design, combined with modern enhancements, ensures that they continue to meet the demands of audio professionals and enthusiasts alike. Recent advancements in microphone technology, particularly in the development of MEMS-based electrodynamic microphones and the exploration of novel materials like graphene, have further enhanced their performance and miniaturization, expanding their use in emerging fields such as wearable devices and advanced acoustic systems. However, challenges like nonlinearities and the optimization of material properties still exist and present opportunities for future research and development.

As microphone technology continues to evolve, the continued refinement of electrodynamic microphones promises to address these challenges while maintaining the advantages that have made them a mainstay in audio recording. Their unique combination of performance, reliability, and versatility ensures that they will remain a cornerstone in the audio industry for years to come.

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